

Linguistic Differences Between Subject Matters Contributing to Tweet Virality

Github: <https://github.com/jcui2001/tweet-topic-virality-analysis>

Research question

Do the linguistic characteristics associated with virality differ across subject matter?

Abstract

This study examines whether linguistic features, author metadata, and topic representations derived from topic modeling can predict tweet virality. Tweets from a publicly available March 2024 Twitter dataset were preprocessed and used to extract features like writing style, readability, sentiment, emotion, and user characteristics. BERTopic was applied to identify discussion topics, which were consolidated into thematic categories for statistical analysis. Mann-Whitney U tests were used to compare sentiment distributions between viral and non-viral tweets within each topic. Logistic Regression, Random Forest, and BERTweet were applied, which were then evaluated using Precision, Recall, F1-score, and ROC-AUC. Model interpretability was examined using Logistic Regression coefficients, Random Forest feature importance, and SHAP values. This work demonstrates how combining topic modeling with interpretable machine learning provides insight into the factors associated with social media engagement. Understanding these patterns can provide insights into how information spreads online and contribute to research on communication and social media behavior.

1. Introduction

Social media plays a major role in how information spreads, yet only a small share of tweets receive substantial engagement. Follower count and topic popularity clearly influence reach, but the language used in a tweet may also contribute to whether it becomes viral. Prior work has used transformer-based models to predict engagement from text, but less attention has been given to whether the linguistic characteristics associated with virality differ across subject matters. This project therefore combines linguistic feature engineering, author metadata, topic modeling, statistical testing, and machine learning to study both prediction and interpretation.

The study addresses three questions:

1. Which linguistic characteristics are associated with tweet virality?
2. Do these relationships differ across discussion topics?

3. How do traditional machine learning models compare with transformer-based language models for predicting virality?

These questions were answered by analyzing tweets using BERTopic, statistical hypothesis testing, and supervised machine learning models.

2. Methodology

2.1 Dataset and Preprocessing

The study used the Twitter New Dataset (March 2024) obtained from Kaggle. Tweets were cleaned by removing special characters, normalizing whitespace, and lemmatizing text. An engagement score was calculated using likes, retweets, and replies, then normalized by follower count to compute an engagement rate. Tweets within the top 10% of engagement rate were labeled as viral, creating a binary classification problem.

2.2 Feature Engineering

To capture multiple aspects of tweet content, several groups of features were extracted:

- Author metadata: follower count, following count, and verification status.
- Linguistic and structural features: character count, word and sentence length, punctuation,
- capitalization, hashtags, mentions, URLs, and question/exclamation marks.
- Readability: Flesch Reading Ease, Flesch-Kincaid Grade Level, and Gunning Fog Index.
- Sentiment and subjectivity: VADER compound sentiment and TextBlob polarity and subjectivity.
- Emotion: transformer-based emotion classification.
- Topic information: manually consolidated subject-matter category.

These features served as inputs for the supervised machine learning models.

2.3 Topic Modeling

BERTopic was applied using OpenAI text-embedding-3-small embeddings to identify semantically related tweet clusters. Representative keywords and sample tweets were manually reviewed to assign descriptive labels. To improve interpretability and increase sample sizes for downstream analyses, the original topics were grouped into broader categories:

- International Affairs & Security
- Government & Public Policy

- Domestic Politics & Society
- Community & Public Services
- Other / Miscellaneous

These five manual categories, rather than the original fine-grained BERTopic clusters, were used for the topic-specific statistical analyses and feature-importance comparisons.

2.4 Statistical Analysis

To determine whether sentiment differed between viral and non-viral tweets, Mann-Whitney U tests were performed within each generalized topic using VADER compound sentiment scores. Violin plots were created to visualize sentiment distributions and complement the statistical tests. A SHAP summary plot of every feature was also produced to understand how different values across the features contribute to the virality of a particular tweet.

2.5 Predictive Modeling

Three supervised learning models were evaluated:

- **Logistic Regression:** Interpretable baseline using engineered features; max_iter = 1000.
- **Random Forest:** Nonlinear model using engineered features; 300 trees.
- **BERTweet:** Transformer pretrained on Twitter text and fine-tuned directly on raw tweet text for 10 epochs with Adam and a learning rate of 2e-5.

Because BERTweet used raw text while Logistic Regression and Random Forest used engineered features and author metadata, the comparison reflects different information inputs as well as different model architectures.

Models were evaluated using Precision, Recall, F1-score, and ROC-AUC.

3. Results

Model	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.2047	0.7909	0.3252	0.7782
Random Forest	0.8750	0.2545	0.4143	0.8857
BERTweet	0.4576	0.2454	0.3195	0.7103

Random Forest achieved the strongest overall performance, with the highest F1-score (0.4143) and ROC-AUC (0.8857). Its precision was very high (0.8750), meaning that tweets labeled viral by the model were usually truly viral, but its recall was lower

(0.2545), so it missed many viral tweets. Logistic Regression showed the opposite trade-off: it achieved high recall (0.7909) but low precision (0.2047), identifying most viral tweets while generating many false positives. BERTweet produced an F1-score of 0.3195 and ROC-AUC of 0.7103. Its weaker performance should not be interpreted as a general failure of transformer models, because it was trained only on tweet text and did not receive the author metadata or engineered features available to the traditional models.

3.1 Topic-Specific Feature Importance

The complete Random Forest feature importance tables for all five subject-matter categories are presented in Appendix A. Below, we summarize the key patterns observed across these topics.

To investigate whether the drivers of virality differed across subject matters, we trained separate Random Forest models for each of the five manually defined topic categories and examined feature importance scores. Although follower count was the most important predictor in every category, the relative importance of linguistic features varied across topics, suggesting that the language associated with viral tweets depends on the subject matter.

Across all five categories, followers consistently ranked as the most influential feature, with the highest importance observed for International Affairs & Security (0.341), followed by Other / Miscellaneous (0.286), Domestic Politics & Society (0.218), Community & Public Services (0.189), and Government & Public Policy (0.162). Following count was also among the most influential predictors in every topic, indicating that author reach and network characteristics are consistently associated with tweet virality.

Beyond author metadata, the importance of linguistic features differed across topics. In Government & Public Policy, subjectivity (0.113) and compound sentiment (0.105) were among the strongest predictors after follower count, suggesting that opinionated and emotionally framed tweets were more strongly associated with engagement in this category. In Community & Public Services, subjectivity, uppercase ratio, question usage, and sentiment all ranked highly, indicating that both writing style and emotional tone contributed to virality. Domestic Politics & Society showed a similar pattern, with uppercase ratio, sentence length, word length, character count, and sentiment all appearing among the top predictors.

For International Affairs & Security, author-level features dominated more strongly than in the other categories, with follower and following counts substantially outweighing linguistic variables. Although writing-style features such as capitalization, subjectivity,

and sentiment still contributed, their relative importance was noticeably lower. Similarly, the Other / Miscellaneous category was influenced by a broader combination of author metadata and linguistic features, with uppercase ratio, average word length, sentiment, character count, and polarity all contributing moderately to prediction performance.

Overall, these findings indicate that while audience size is consistently the strongest determinant of virality, the linguistic characteristics associated with highly engaging tweets are not uniform across subject matters. Instead, different topics emphasize different combinations of sentiment, writing style, and text structure, supporting our hypothesis that the relationship between language and virality is topic dependent.

3.2 Overall Interpretation

Taken together, the findings indicate that follower count is consistently important but it does not fully explain virality. Linguistic characteristics also contribute, and their relative importance changes by subject matter. The results therefore support a topic-specific interpretation rather than a single universal set of language features that predicts engagement equally well across all tweets.

4. Discussion

This study demonstrates that tweet virality can be analyzed through a combination of linguistic feature engineering, topic modeling, and machine learning. BERTopic provided meaningful thematic groupings that enabled topic-specific statistical analysis, while feature importance methods identified the characteristics most strongly associated with engagement.

The comparison of Logistic Regression, Random Forest, and BERTweet illustrates the trade-off between interpretability and predictive performance. Traditional models provide transparent explanations through coefficients and feature importance, whereas BERTweet learns contextual representations directly from tweet text. Together, these approaches offer both predictive capability and insight into the factors influencing social media engagement.

5. Limitations and Future Work

- The dataset contains English-language tweets from a single month, limiting temporal and linguistic generalizability.
- Virality was defined using the top 10% of follower-normalized engagement; results may differ under other thresholds or alternative engagement formulas.
- The analysis did not include multimedia content, posting time, social-network structure, or diffusion pathways.
- The five subject matters were manually consolidated from BERTopic clusters, which improves interpretability but introduces researcher judgment.

- BERTweet did not receive the same metadata and engineered inputs as the traditional models, so the architecture comparison was not fully controlled.
- Some topic-specific feature tables remain incomplete in the current report; future work should present the same number of ranked features for all five subject matters.
- Future models could combine BERTweet text embeddings with author metadata and engineered features, use class weighting or resampling, and evaluate results with cross-validation and confidence intervals.

6. Conclusion

This project developed a framework for predicting Twitter virality using linguistic features, author metadata, BERTopic, five manually consolidated subject matters, and supervised machine learning. Random Forest achieved the best overall F1-score and ROC-AUC, while Logistic Regression achieved the highest recall. Most importantly, the topic-specific analyses showed that the features associated with virality are not identical across subject matters. Follower count remained highly influential, but subjectivity, capitalization, question use, sentiment, and text-length measures also contributed in topic-dependent ways. These findings support the conclusion that understanding virality requires considering both who posts a tweet and how the tweet is written within its subject-matter context.

7. Author Contributions

Hanako Rahdert: Data cleaning and preprocessing; virality metric definition; BERTopic topic modeling with OpenAI embeddings; topic interpretation and labeling; presentation slides; report revision.

Jeremy Cui: Topic modeling, feature engineering, feature enrichment, statistical analysis, classification and ensemble modeling (Logistic regression, Random forest), transformer modeling, presentation slide.

References

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Appendix A

Community & Public Services

Feature	Importance
followers	0.189156
following	0.161031
subjectivity	0.076875
uppercase_ratio	0.071402
num_question	0.067366
sent_compound	0.062036
char_count	0.061492
avg_word_length	0.060335
polarity	0.045455
avg_sentence_length	0.042365

Domestic Politics & Society

Feature	Importance
followers	0.217960
following	0.161068
uppercase_ratio	0.073975
avg_sentence_length	0.064797
avg_word_length	0.063991
char_count	0.062051
sent_compound	0.057031
polarity	0.055599
word_count	0.051480
subjectivity	0.044791

Government & Public Policy

Feature	Importance
followers	0.162012
subjectivity	0.113148
sent_compound	0.105362
char_count	0.093752
uppercase_ratio	0.082601
polarity	0.071582
following	0.068934
avg_sentence_length	0.059312
word_count	0.054112
avg_word_length	0.063991

International Affairs & Security

Feature	Importance
followers	0.341338
following	0.164693

Appendix A

uppercase_ratio	0.055741
char_count	0.049494
subjectivity	0.047262
sent_compound	0.043053
avg_word_length	0.040694
word_count	0.039258
avg_sentence_length	0.032030
polarity	0.029439

Other / Miscellaneous

Feature	Importance
followers	0.285728
following	0.115712
uppercase_ratio	0.074156
avg_word_length	0.064575
sent_compound	0.054899
char_count	0.054221
word_count	0.048496
polarity	0.048220
subjectivity	0.045538
avg_sentence_length	0.044371